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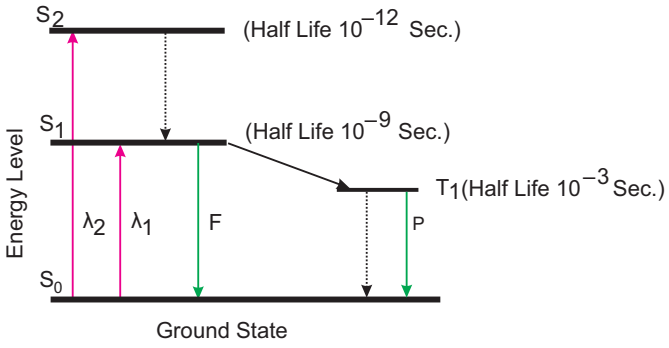


Figure 13.11: Fluorescence (F) and Phosphorescence (P)

radiations (Figure 13.11). Pathway of electron during fluorescence: $S_1 \rightarrow S_0$

ii. Phosphorescence

Electron from Second Singlet State (S_2) may return to next higher energy level (S_1) by losing some of its extra energy in the form of heat. From first singlet state (S_1) electron further drops to first triplet state (T_1). Triplet State is unstable having half life time of 10^{-3} seconds and electrons returns to ground state with emission of light in red region called as **phosphorescence** (Figure 13.11). Phosphorescence is the delayed emission of absorbed radiations. Pathway of electron during Phosphorescence: $S_2 \rightarrow S_1 \rightarrow T_1 \rightarrow S_0$

13.9.2 Photosystem and Reaction Centre

- Thylakoid membrane contains Photosystem I (PS I) and Photosystem II (PS II).
- PS I is in unstacked region of granum facing stroma of chloroplast.

CPR3BP

- PS II is found in stacked region of thylakoid membrane facing lumen of thylakoid.
- Each Photosystem consists of central core complex (CC) and light harvesting Complex (LHC) or Antenna molecules(Figure 13.12).
- The core complex consists of respective reaction centre associated with proteins, electron donors and acceptors.
- PS I – CC I consists of reaction centre P700 and LHC I.

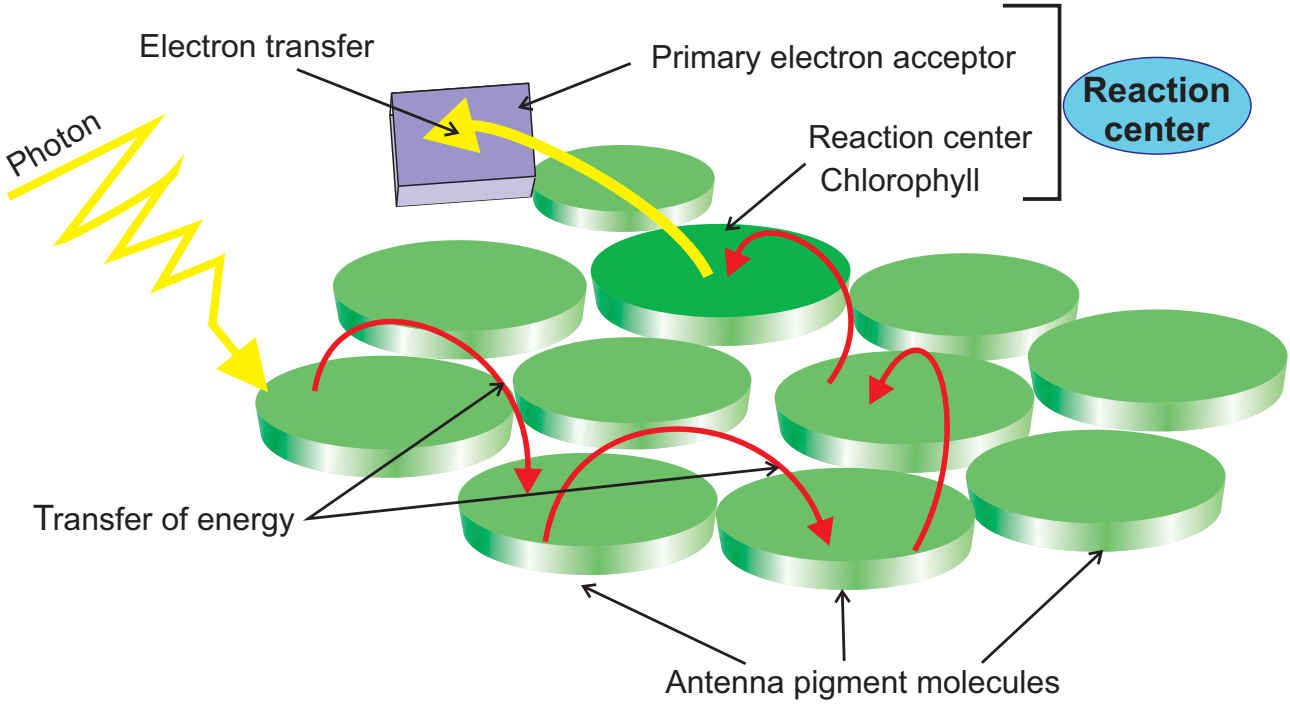


Figure 13.12: Photosystem